WEEK 10: REAL-ESTATE ANALYTICS DASHBOARD

Aim:

To create an interactive real-estate analytics dashboard in Power BI to analyze property prices, locations, characteristics, and sales patterns.

Procedure

Step 1: Open the Week 9 Project

1. Open Power BI Desktop.
2. Open the saved Week 9 project.

Step 2: Create the Total Properties Sold Measure

1. Create a new table for measures.
2. Click **Modeling** → **New Measure**.
3. Enter the formula:

```
Total Properties Sold = COUNTROWS('HouseSales')
```

4. Press **Enter**.

Step 3: Create the Average House Price Measure

1. Click **Modeling** → **New Measure**.
2. Enter:

```
Avg House Price = AVERAGE('HouseSales'[price])
```

3. Press **Enter**.

Step 4: Create the Median Price Measure

1. Click **Modeling** → **New Measure**.
2. Enter:

```
Median Price = MEDIAN('HouseSales'[price])
```

3. Press **Enter**.

Step 5: Create the Average Living Area Measure

1. Click **Modeling** → **New Measure**.
2. Enter:

```
Avg Living Area = AVERAGE('HouseSales'[sqft_living])
```

3. Press **Enter**.

Step 6: Create the Average Price per SqFt Measure

1. Click **Modeling** → **New Measure**.
2. Enter:

   Avg Price per SqFt = AVERAGE('HouseSales'[Price per SqFt])
3. Press **Enter**.
4. Format the price measures as Currency and the count and living-area measures as Whole Number where appropriate.

Step 7: Display the Measures Using Cards

1. Open **Report view**.
2. Select the Card visual.
3. Drag Total Properties Sold into the card.
4. Repeat the procedure to create separate cards for Avg House Price, Median Price, Avg Living Area, and Avg Price per SqFt.
5. Arrange the five cards at the top of the report page.

Step 8: Create the Geographic Map

1. Select an empty area on the report canvas and Select the Map visual.
2. Drag lat into Latitude and long into Longitude.
3. Drag Price Tier into Legend.
4. Drag price into Size if the visual supports that field.
5. Add price, bedrooms, bathrooms, sqft_living, and grade to Tooltips.
6. Set the data categories of lat and long to Latitude and Longitude respectively.
7. Adjust the map size and check that property locations appear.

Step 9: Create the Average Price by Bedrooms Chart

1. Select a Clustered Column Chart.
2. Drag bedrooms into the X-axis.
3. Drag price into the Y-axis.
4. Change the aggregation of price to Average.
5. Rename the chart title to **Average House Price by Bedrooms**.

Step 10: Create the Average Price by Bathrooms Chart

1. Add another Clustered Column Chart.
2. Drag bathrooms into the X-axis.
3. Drag price into the Y-axis and set the aggregation to Average.
4. Set the title to **Average House Price by Bathrooms**.

Step 11: Create the Average Price by Waterfront Chart

1. Add another Clustered Column Chart.
2. Drag waterfront into the X-axis.
3. Drag price into the Y-axis and set the aggregation to Average.
4. Set the title to **Average House Price by Waterfront Status**.

Step 12: Create the Grade and Condition Charts

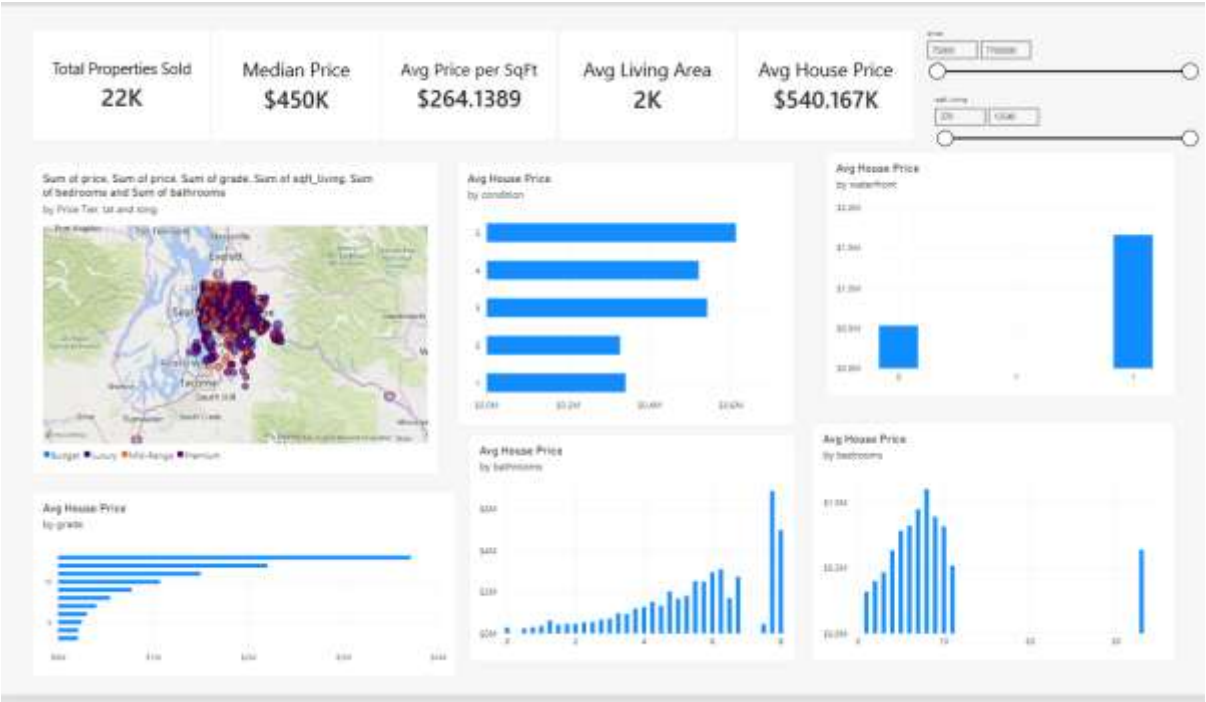
1. Add a Clustered Bar Chart.
2. Drag grade into the Y-axis and price into the X-axis.
3. Set the aggregation of price to Average.
4. Name the chart **Average House Price by Grade**.
5. Repeat the procedure using condition on the Y-axis.
6. Name the second chart **Average House Price by Condition**.

Step 13: Create the Living-Area Range Slicer

1. Select the Slicer visual.
2. Drag sqft_living into its field well.
3. Set the minimum and maximum square-footage values to filter the properties.
4. Check that the other visuals respond to the selected range.

Step 14: Create the Price Range Slicer

1. Add another Slicer visual.
2. Drag price into its field well.
3. Select the required minimum and maximum prices.
4. Test the slicer to confirm that the dashboard updates.



Result:

An interactive House Sales Analytics Dashboard was created in Power BI. It displays the main property statistics, geographic distribution, price comparisons by property characteristics, and dynamic range filters for exploring housing data.